EDA_titanic

April 30, 2025

[1]:
```
import pandas as pd

df = pd.read_csv('titanic.csv')
```

[2]:
```
df.head()
df.info()
df.describe()
df.isnull().sum()
df['Survived'].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Survived     418 non-null    int64
 2   Pclass       418 non-null    int64
 3   Name         418 non-null    object
 4   Sex          418 non-null    object
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64
 7   Parch        418 non-null    int64
 8   Ticket       418 non-null    object
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object
 11  Embarked     418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

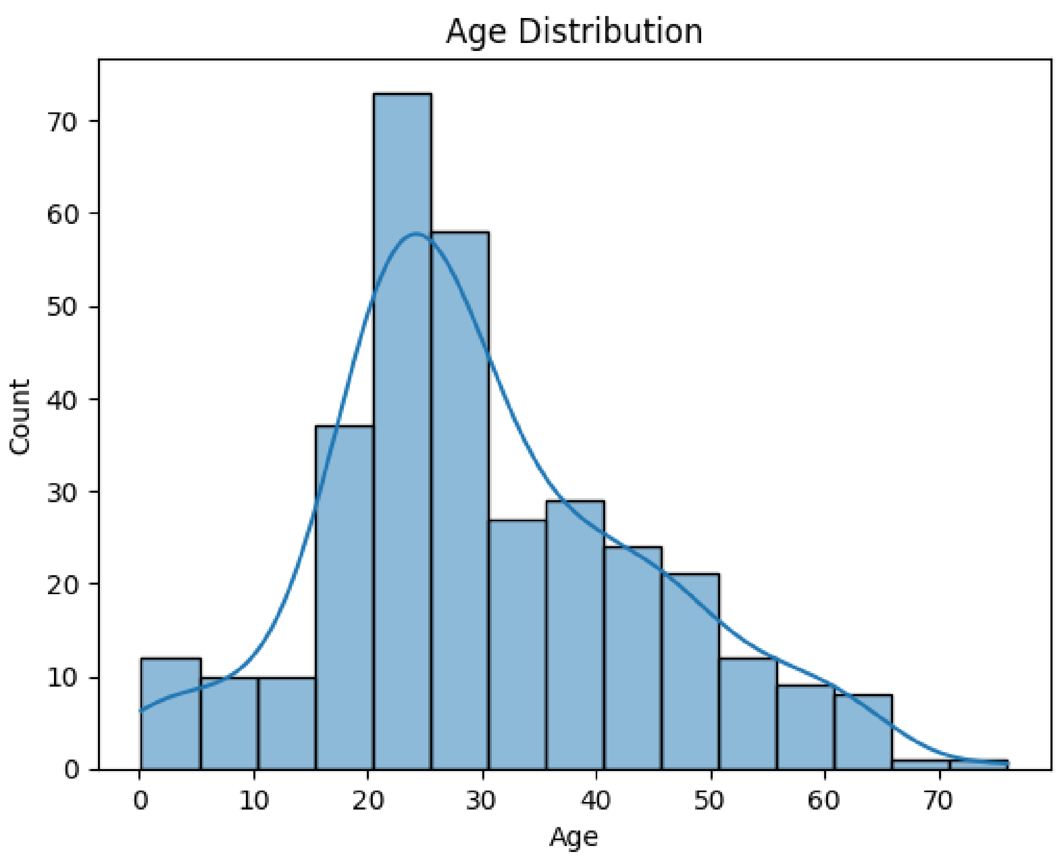
[2]:
```
Survived
0    266
1    152
Name: count, dtype: int64
```

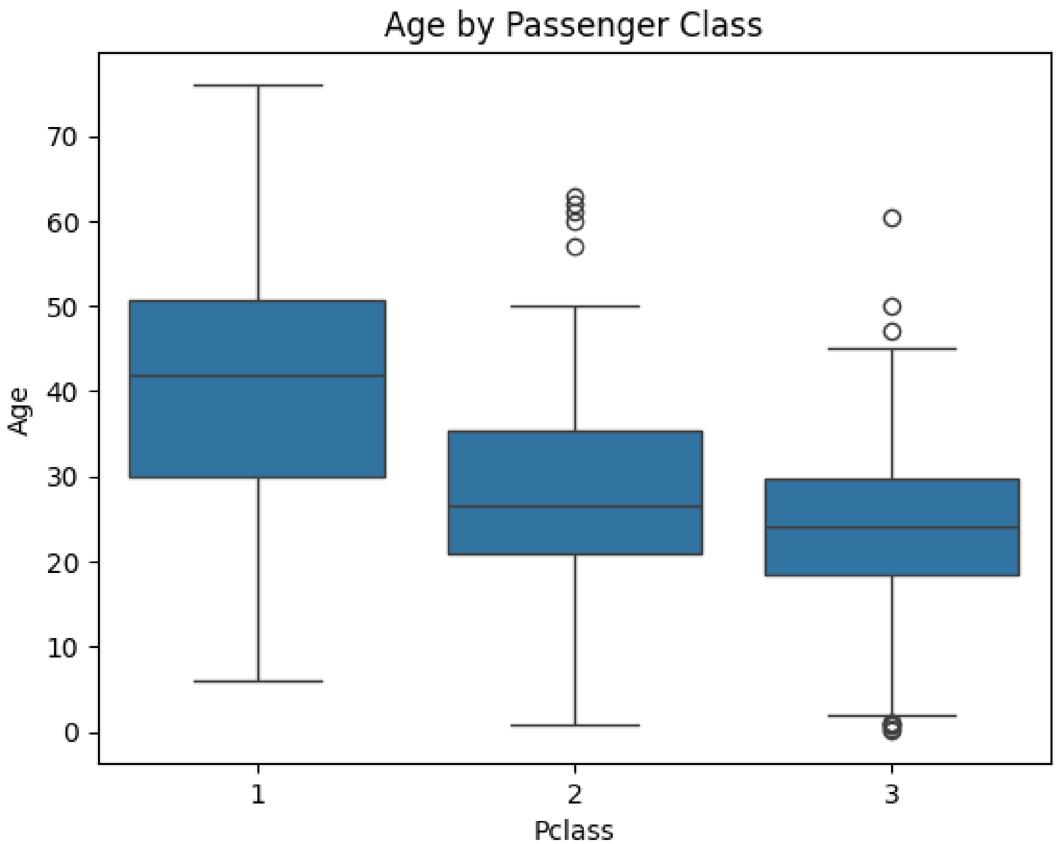
[7]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

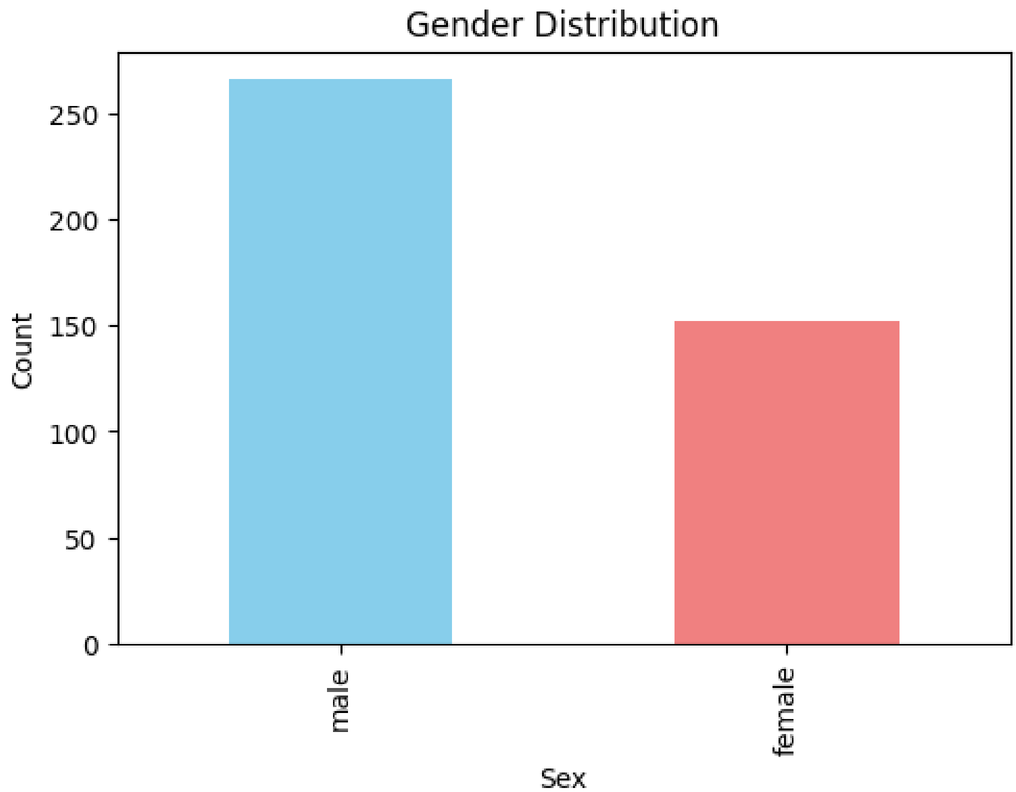
```
sns.histplot(df['Age'], kde=True)
plt.title('Age Distribution')
plt.show()

# Observation: Age is right-skewed. Most passengers are between 20 and 40.
sns.boxplot(x='Pclass', y='Age', data=df)
plt.title('Age by Passenger Class')
plt.show()

# Observation: 1st class passengers are generally older; 3rd class includes␣
↪more younger individuals.
plt.figure(figsize=(6, 4))
df['Sex'].value_counts().plot(kind='bar', color=['skyblue', 'lightcoral'])
plt.title('Gender Distribution')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.show()
```




Age by Passenger Class
Age
70
60
50
40
30
20
10
0
1
2
3
Pclass

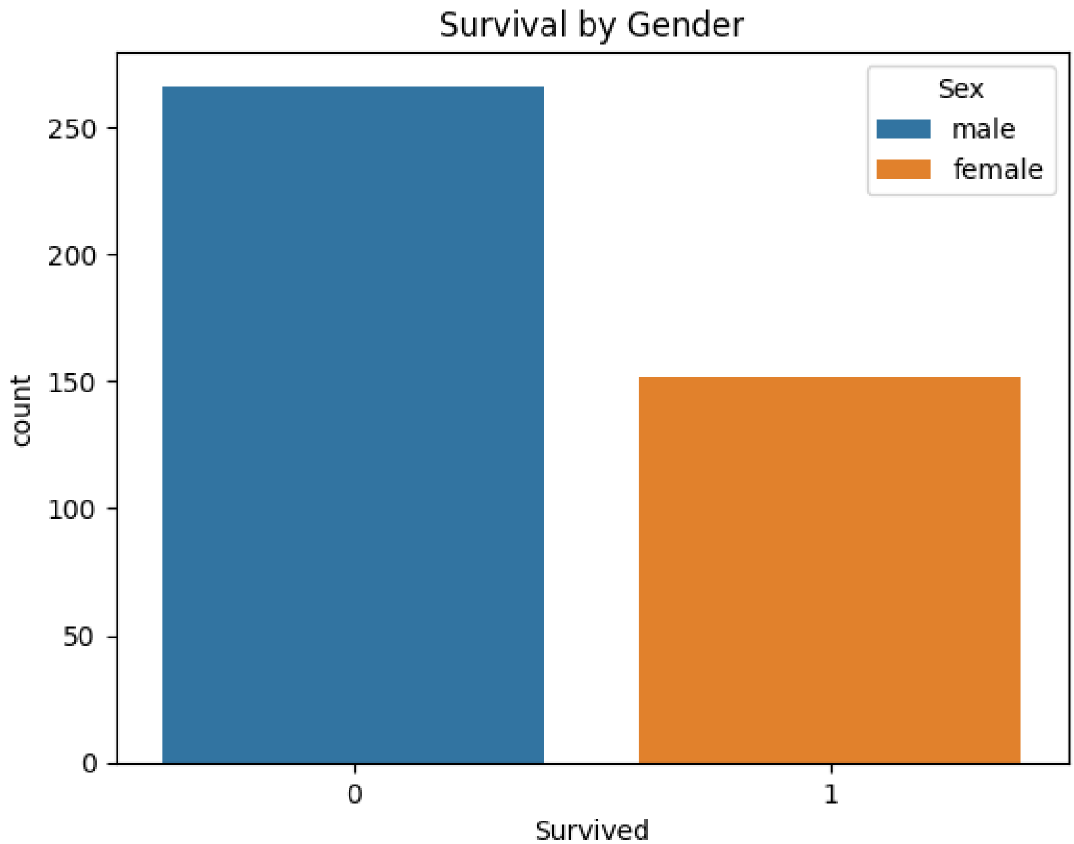


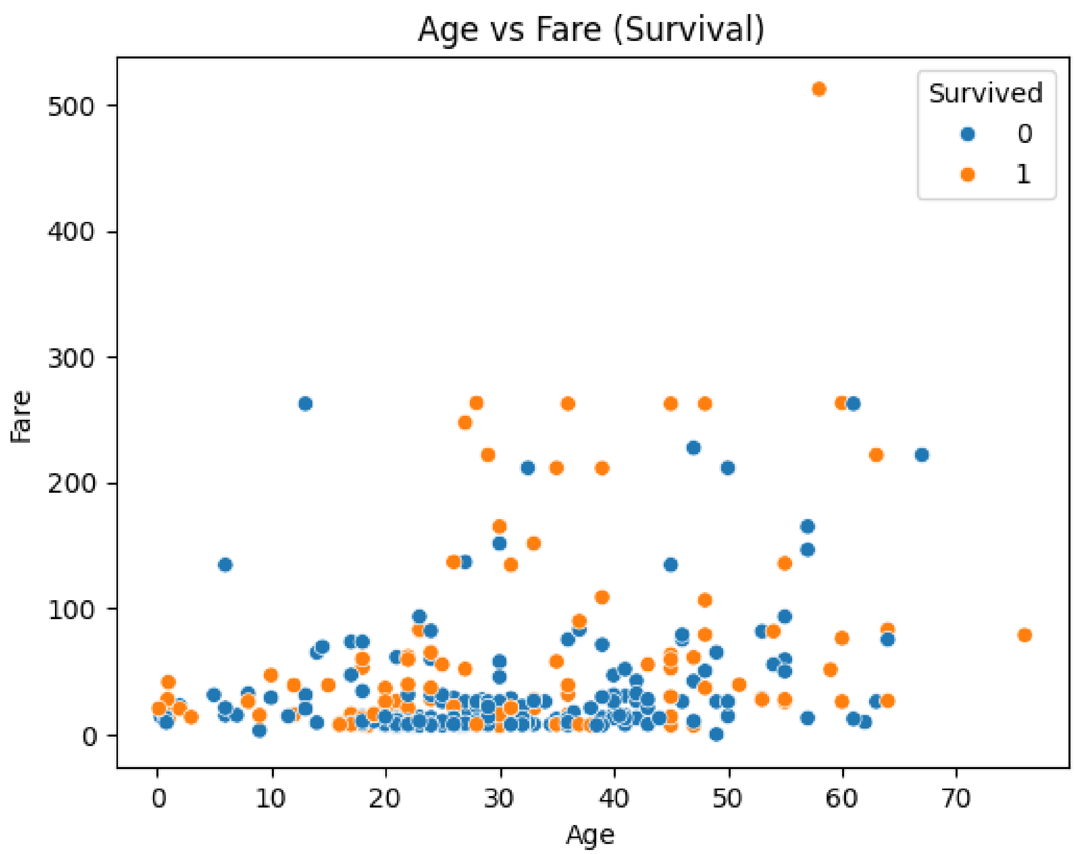
Women were more likely to survive.

Higher class (1st class) had higher survival.

[ ]:
```
# Observation: There are more males than females on board.
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title('Survival by Gender')
plt.show()

# Observation: Female passengers had a much higher survival rate than males.
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title('Age vs Fare (Survival)')
plt.show()
```

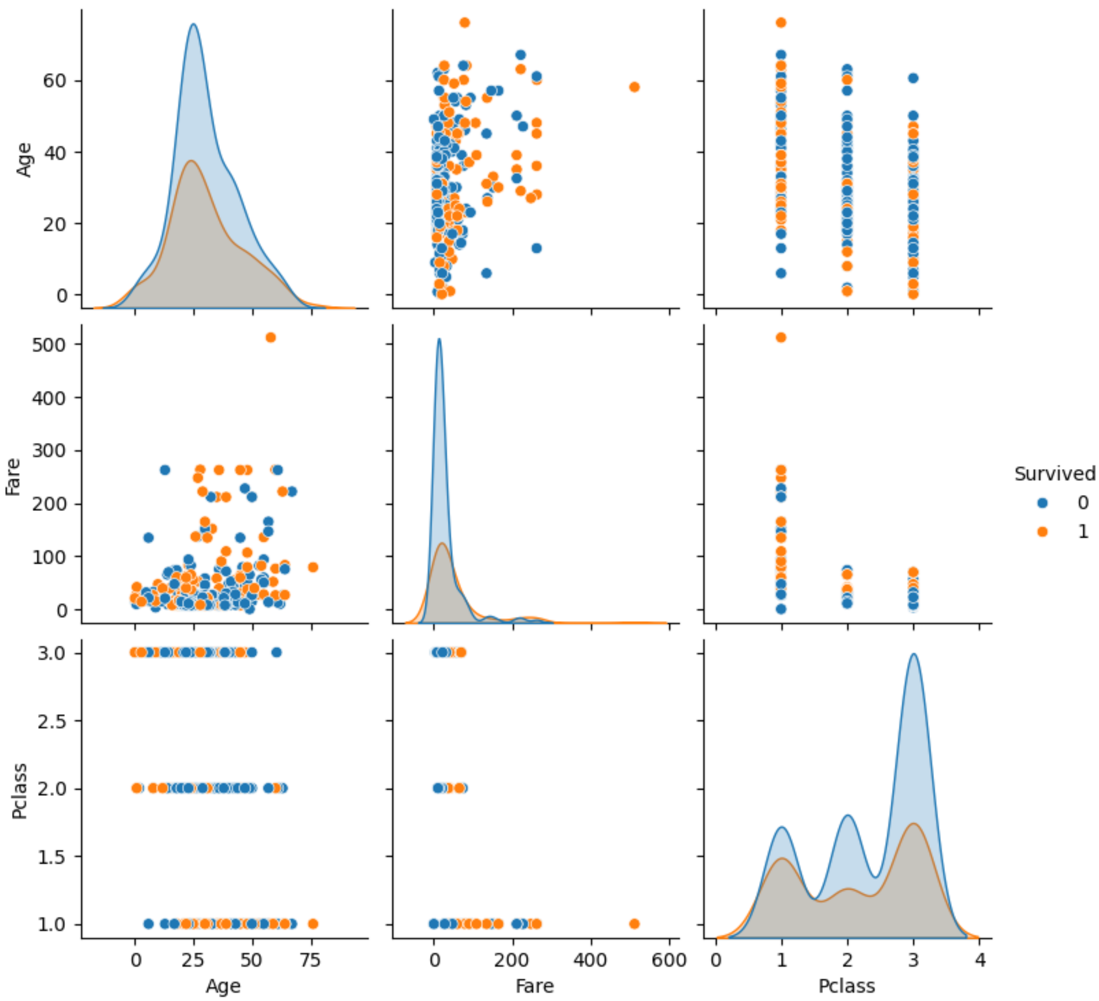

Survival by Gender
Sex
male
female
count
250
200
150
100
50
0
0
1
Survived

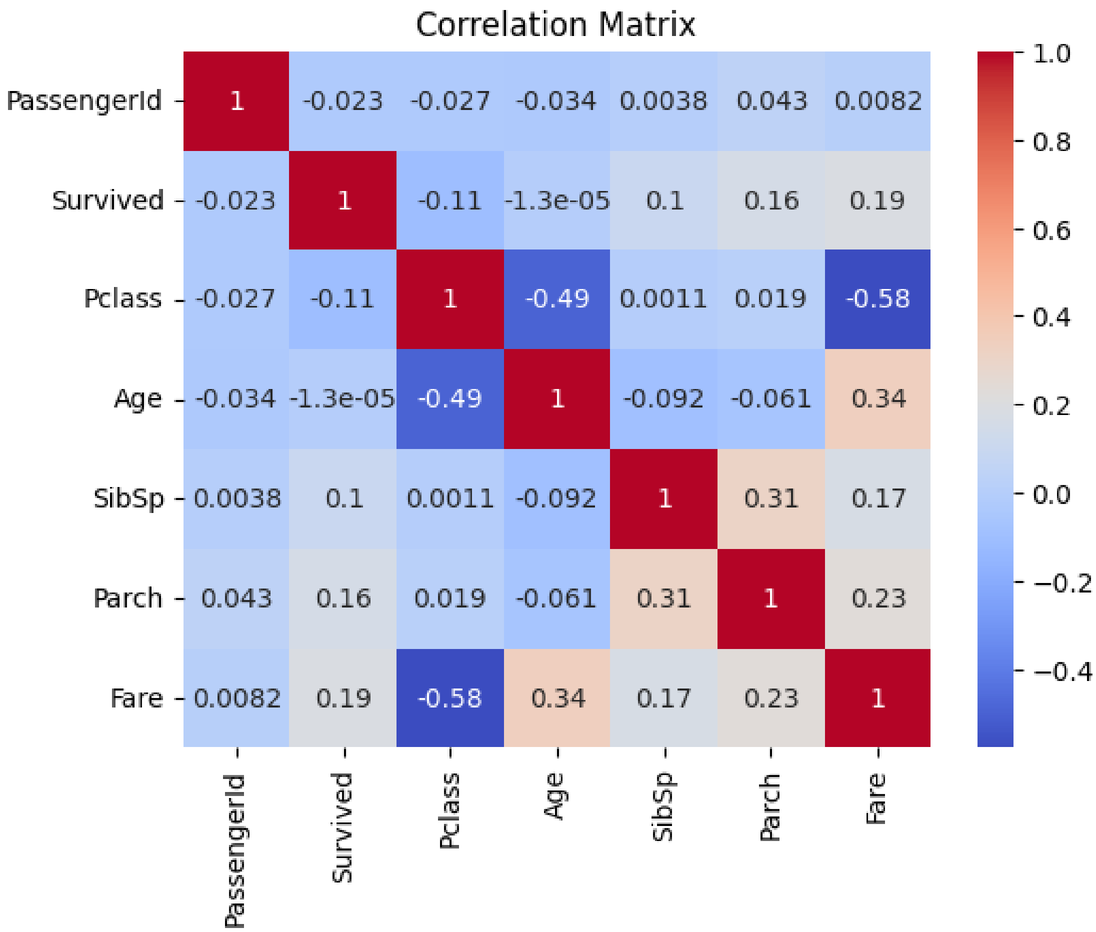


Strong negative correlation between Pclass and Survived

```
[ ]: #  Multivariate Analysis
# Observation: Higher fares and younger ages are loosely associated with higher␣
 ↪survival.
sns.pairplot(df[['Age', 'Fare', 'Survived', 'Pclass']], hue='Survived')
plt.show()

# Observation: Survivors tend to be in higher classes and pay higher fares.␣
 ↪Some separation is visible.
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```


Age
Fare
Pclass
Survived
0
1
0
20
40
60
100
200
300
400
500
1.0
1.5
2.0
2.5
3.0
25
50
75
600
3
4



Observation: Survived is moderately correlated with Fare and negatively with Pclass. Age has a weaker correlation.

0.1 Summary

- Female passengers had a much higher survival rate.
- 1st class passengers were more likely to survive.
- Most passengers were between 20-40 years old.
- Age and Cabin have missing values that should be handled before modeling.
- Fare is positively correlated with survival, while Pclass is negatively correlated.